

**Time: 2:30 Hours**

**Total Marks-50**

=====

**Answer the following questions; Figures in the right hand margin indicate full marks**

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1. a) How to check if pain receptors are active in the human body? Give an example. 2  
b) What is Finger Print ? Explain briefly 2  
c) What does Allah Almighty say in the Quran about fingerprint? 2  
d) Highlight the ethical issues of computer use from the perspective of Islam. 4
  
2. a) Explain the physical and spiritual benefits of Salat and Fasting (Sawom) 6  
b) Write a brief summary of 4 Muslim Scientists mentioning their contributions to their respective fields 4
  
3. What are the negative impacts of having sex outside of marriage, free mixing, extramarital affair, homosexuality, pornography and short dress on society? 10
  
4. a) Classify disasters with examples 2  
b) Mention three examples of natural disasters described in the glorious Qur'an 3  
c) The Messenger of Allah said Natural Disaster occurs due to fourteen crimes. What are those? 5  
**Or**  
a) What is Miracle (Mojeza)? Mention four different miracles by four different messengers by mentioning Quranic verses. 5  
b) The Qur'an itself is a miracle "- explain this statement with your logical argument 5
  
5. a) What do you understand by Teaching and Effective Teaching? Explain how Learning Process Works? 5  
b) How every prophets including Muhammad (PBUH) played roles as teachers? 5  
**Or**  
a) What is autophagy? 2  
b) How does autophagy work? 4  
c) Who was awarded the Nobel Prize in 2016 by the Nobel Committee for the discovery of autophagy? 2  
d) Our body contains the hormone IGF (IGF-Insulin Growth Factor). What is the function of this hormone? 2

**International Islamic University Chittagong**  
**Center for General Education (CGED)**

**Semester End-Examination: Autumn 2024**  
**Course Code: GEHE-3601**

**Program: Undergraduate**

**Course Title: History of Emergence**  
**Bangladesh**  
**Full Marks: 50**

**Time: 2 hours and 30 minutes.**

**Instructions:**

- i. All Questions are Compulsory.
- ii. Figures in the right margin indicate full marks.
- iii. Course Learning Outcome (CLO) and Bloom's levels are mentioned in additional columns.

**Bloom's Levels of the Questions.**

| Letter of Symbol | R        | U          | App   | An      | E        |
|------------------|----------|------------|-------|---------|----------|
| Meaning          | Remember | Understand | Apply | Analyze | Evaluate |

|   | Text of the Questions                                                                                                                                                                                                                                                                                                                                              | Marks  | Bloom's Level | CLO  |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------|------|
| 1 | Evaluate the different phases of the Language Movement of 1952 with its significances.                                                                                                                                                                                                                                                                             | 10     | E             | CLO2 |
| 2 | Assess the historical context of the formation of the United Front and investigate the key factors of its success in the election of 1954 highlighting major points of its election manifesto.<br>Or<br>Analyze the historical background of the Six-point Demands of 1966. How did the demands play an important role in the emergence of independent Bangladesh? | 10     | An            | CLO3 |
| 3 | Appreciate the background of the Election 1970 and evaluate the result of the Election with its influence in the Independence War of 1971.                                                                                                                                                                                                                         | 10     | E             | CLO3 |
| 4 | Explain the responses and role of superpowers in the Liberation War of Bangladesh in the light of contemporary Cold War.                                                                                                                                                                                                                                           | 10     | E             | CLO3 |
| 5 | Write short notes on any two:<br>a) Operation Searchlight<br>b) Agartala Conspiracy Case of 1968<br>c) Eleven Sectors during the Liberation War of 1971.                                                                                                                                                                                                           | 5<br>5 | E             | CLO3 |



# International Islamic University Chittagong

Department of Computer Science and Engineering

Final Exam, Autumn 2024

Course Code: ECON-3501

Program: B. Sc. Engg

Course Title: Principles of Economics

Time: 2 hours 30 minutes

Full Marks: 50

(i) Answer all the questions. The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

| Course Outcomes (COs) of the Questions |                                                |
|----------------------------------------|------------------------------------------------|
| CO1                                    | Explain the fundamental concepts of Economics. |
| CO2                                    | Analyze the key indicators of economic growth. |
| CO3                                    | Compare the economic theories and concepts.    |

| Bloom's Levels of the Questions |          |            |       |         |          |        |
|---------------------------------|----------|------------|-------|---------|----------|--------|
| Letter Symbols                  | R        | U          | App   | An      | E        | C      |
| Meaning                         | Remember | Understand | Apply | Analyze | Evaluate | Create |

| Part A<br>[Answer the following questions ] |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks              | CLO       | DL              |                    |    |      |       |    |    |      |       |    |    |      |       |     |    |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |   |      |   |
|---------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------|-----------------|--------------------|----|------|-------|----|----|------|-------|----|----|------|-------|-----|----|------|-------|-----|-----|------|-------|-----|-----|------|-------|-----|-----|------|-------|-----|-----|------|-------|-----|-----|------|-------|-----|-----|------|-------|-----|---|------|---|
| 1                                           | a)        | Explain monopoly and monopolistic competition market structure, and identify the key factors that distinguish them.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 5                  | CLO3      | An              |                    |    |      |       |    |    |      |       |    |    |      |       |     |    |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |   |      |   |
| 1.                                          | b)        | <p>The table below shows the market demand schedule and the cost structure.</p> <table><tr><th>Quantity</th><th>Price per</th><th>Fixed cost (FC)</th><th>Variable cost (VC)</th></tr><tr><td>20</td><td>\$15</td><td>\$200</td><td>40</td></tr><tr><td>35</td><td>\$15</td><td>\$200</td><td>70</td></tr><tr><td>55</td><td>\$15</td><td>\$200</td><td>110</td></tr><tr><td>75</td><td>\$15</td><td>\$200</td><td>150</td></tr><tr><td>105</td><td>\$15</td><td>\$200</td><td>210</td></tr><tr><td>130</td><td>\$15</td><td>\$200</td><td>260</td></tr><tr><td>160</td><td>\$15</td><td>\$200</td><td>320</td></tr><tr><td>190</td><td>\$15</td><td>\$200</td><td>380</td></tr><tr><td>220</td><td>\$15</td><td>\$200</td><td>440</td></tr><tr><td>250</td><td>\$15</td><td>\$200</td><td>500</td></tr></table> <p>Measure Total cost (TC), Average total cost (ATC) and Marginal cost (MC) for each row.</p> | Quantity           | Price per | Fixed cost (FC) | Variable cost (VC) | 20 | \$15 | \$200 | 40 | 35 | \$15 | \$200 | 70 | 55 | \$15 | \$200 | 110 | 75 | \$15 | \$200 | 150 | 105 | \$15 | \$200 | 210 | 130 | \$15 | \$200 | 260 | 160 | \$15 | \$200 | 320 | 190 | \$15 | \$200 | 380 | 220 | \$15 | \$200 | 440 | 250 | \$15 | \$200 | 500 | 5 | CLO2 | U |
| Quantity                                    | Price per | Fixed cost (FC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Variable cost (VC) |           |                 |                    |    |      |       |    |    |      |       |    |    |      |       |     |    |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |   |      |   |
| 20                                          | \$15      | \$200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 40                 |           |                 |                    |    |      |       |    |    |      |       |    |    |      |       |     |    |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |   |      |   |
| 35                                          | \$15      | \$200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 70                 |           |                 |                    |    |      |       |    |    |      |       |    |    |      |       |     |    |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |   |      |   |
| 55                                          | \$15      | \$200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 110                |           |                 |                    |    |      |       |    |    |      |       |    |    |      |       |     |    |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |   |      |   |
| 75                                          | \$15      | \$200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 150                |           |                 |                    |    |      |       |    |    |      |       |    |    |      |       |     |    |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |   |      |   |
| 105                                         | \$15      | \$200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 210                |           |                 |                    |    |      |       |    |    |      |       |    |    |      |       |     |    |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |   |      |   |
| 130                                         | \$15      | \$200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 260                |           |                 |                    |    |      |       |    |    |      |       |    |    |      |       |     |    |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |   |      |   |
| 160                                         | \$15      | \$200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 320                |           |                 |                    |    |      |       |    |    |      |       |    |    |      |       |     |    |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |   |      |   |
| 190                                         | \$15      | \$200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 380                |           |                 |                    |    |      |       |    |    |      |       |    |    |      |       |     |    |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |   |      |   |
| 220                                         | \$15      | \$200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 440                |           |                 |                    |    |      |       |    |    |      |       |    |    |      |       |     |    |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |   |      |   |
| 250                                         | \$15      | \$200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 500                |           |                 |                    |    |      |       |    |    |      |       |    |    |      |       |     |    |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |   |      |   |
| 2.                                          | a)        | <p>Differentiate the following with example:</p> <p>i. Long run cost and Short run cost.</p> <p>ii. Monopoly and Monopolistic competition market</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 5                  | CLO1      | U               |                    |    |      |       |    |    |      |       |    |    |      |       |     |    |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |   |      |   |
| 2.                                          | b)        | <p>If the Given the total cost function <math>TC = 600 + 20Q - 6Q^2 + Q^3</math> Where TC is total cost and Q is level of output.</p> <p>Calculate Total cost, ATC, AVC and MC when the firm produces 350 units of output.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 5                  | CLO2      | An              |                    |    |      |       |    |    |      |       |    |    |      |       |     |    |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |     |      |       |     |   |      |   |

Or

|                                                                                       |    |                                                                                                                                                            |   |      |    |
|---------------------------------------------------------------------------------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|----|
| 2.                                                                                    | a) | What is Inflation in Economics? What are the causes of Inflation? Discuss briefly.                                                                         | 5 | CLO1 | U  |
| 2.                                                                                    | b) | Describe about the 'Vicious Circle of Poverty'. How we can overcome this vicious circle?                                                                   | 5 | CLO2 | An |
| <p style="text-align: center;"><b>Part B</b><br/>[Answer the following questions]</p> |    |                                                                                                                                                            |   |      |    |
| 3.                                                                                    | a) | Discuss monetary policy and fiscal policy. How fiscal policy plays an important role in a developing country like Bangladesh?                              | 5 | CLO2 | U  |
| 3.                                                                                    | b) | Briefly explain unemployment with its classification. Evaluate the measures that can be taken to reduce unemployment problem in a country like Bangladesh. | 5 | CLO2 | An |
| 4.                                                                                    | a) | How do government subsidies help an industry? Explain.                                                                                                     | 5 | CLO1 | U  |
| 4.                                                                                    | b) | What is the impact of the coronavirus pandemic on international trade?                                                                                     | 5 | CLO2 | An |
| 5.                                                                                    | a) | Evaluate the relationship between Technological progress and economic growth in Bangladesh.                                                                | 5 | CLO2 | An |
|                                                                                       | b) | Explain the difference between growth and development?                                                                                                     | 5 | CLO2 | An |
| <p style="text-align: center;">Or,</p>                                                |    |                                                                                                                                                            |   |      |    |
| 5.                                                                                    | a) | Distinguish between Economic Growth and Economic Development.                                                                                              | 5 | CLO2 | An |
|                                                                                       | b) | Describe some obstacles of economic development.                                                                                                           | 5 | CLO2 | An |



**International Islamic University Chittagong**  
**Center for General Education (CGED)**  
**Semester End Examination- Autumn- 24**

**Course Code: URED- 3604 (URED-3102 for LLB)**

**Course Title: Life and Teachings of the Prophet (SAAS)**

**Time: 2:30 Hours**

**Full Marks: 50**

*Answer the following questions  
(All questions are of equal value)*

| SL | Questions                                                                                                                                                                                                                                    | Marks | CLO | Bloom's taxonomy domain    |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|----------------------------|
| 1  | Discuss the background of the <i>Hijrah</i> of Prophet (SAAS) focusing on its significance as "the turning point" in the history of Islam.                                                                                                   | 10    | 3   | Understand                 |
| 2  | Explain the background of battle of Badr. Evaluate the everlasting impact of Battle of Badr as the decisive battle between the truth and wrong.                                                                                              | 10    | 3   | Create & Evaluate          |
| 3  | Analyze the background of battle of Ahzab assessing the strategies of the Prophet Muhammad (SAAS) for the success in this battle.                                                                                                            | 10    | 3   | Analyze & Evaluate         |
| 4  | Discuss the background of the victory of Makkah. Explain how the teaching of the victory of Makkah should be implemented in the political and social context of Bangladesh.                                                                  | 10    | 3   | Understand & Create        |
| 5  | Evaluate the Farewell Address of Prophet Muhammad (SAAS) as an acknowledged document of human rights.<br>Or<br>Write short notes on any two of the following:<br>a) Charter of Madinah<br>b) Covenant of Aqabah<br>c) Treaty of Hudaibiyyah. | 10    | 3   | Evaluate<br><br>Understand |

# International Islamic University Chittagong

Department of Computer Science & Engineering  
B.Sc. in CSE, Final Examination, Autumn 2024

Course Code: CSE-3525

Total Marks: 50

Course Title: Data Communication

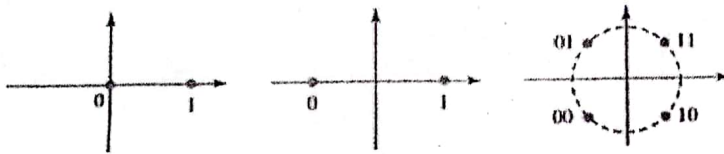
Time: 2 Hours 30 Minutes

[Answer all the following questions. Figures in the right hand margin indicate full marks. Use a Separate answer script for Group-A and Group-B.]

COs

## Group A

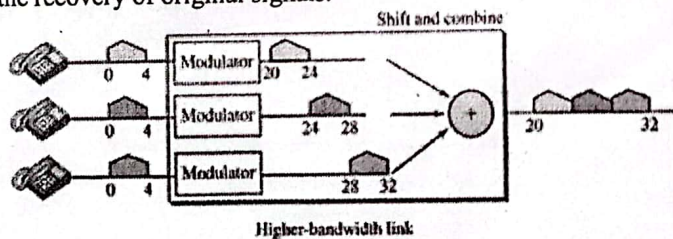
1. a) We need to send data 10 bits at a time at a bit rate of 10 Mbps using FSK. The carrier frequency is 10 MHz. Calculate the number of levels, baud rate and bandwidth. Show the allocation of frequencies and bandwidth. 5 An 3
- b) Write differences between Amplitude Modulation and Frequency Modulation. Calculate the baud rate for the given bit rate and type of modulation. 5 An 3
  - a. 2000 bps, FSK
  - b. 4000 bps, ASK
  - c. 6000 bps, QPSK
  - d. 36,000 bps, 64-QAM
2. a) Explain the number of signal elements and data elements in each of the following figures: 5 U 3



- b) Explain Frequency Modulation with figure. Why guard bands are required between FM radio stations? 5 U 1
- OR
2. a) Using data "10111001" draw the waveform for BFSK and BPSK techniques. Assume these bits travels in 1 second. 5 U 3
  - b) How many FM radio channel can be operated at a time in a an area? FM radio operates in the 88 MHz to 108 MHz frequency band. 5 U 1

## Group B

3. a) What type of multiplexing is given in the figure? Draw the diagram at the receiver side showing the recovery of original signals. 5 U 4



- b) Explain synchronous TDM with figure, what are it's advantages? 5 U 4
- OR
3. a) Assume seven channels, each with a 100-kHz bandwidth, are to be multiplexed together using FDM. What is the minimum bandwidth of the link if there is a need for a guard band of 20 kHz between the channels? Why guard band is needed here explain with appropriate figure. 5 U 4



- b) How does Statistical TDM differ from Traditional Time Division Multiplexing (TDM). If four 1-kbps connections are multiplexed together. A unit is 1 bit. Find--

- Find the duration of 1 bit before multiplexing,
- Find the transmission rate of the link
- Find the duration of a time slot
- Find the duration of a frame
- Find the frame rate

4. a) Show Byte Stuffing and unstuffing for the following data:

|      |      |      |     |     |     |  |
|------|------|------|-----|-----|-----|--|
| Flag | Flag | Flag | ESC | ESC | ESC |  |
|------|------|------|-----|-----|-----|--|

What is bit stuffing?

- b) Assume a sender sends 4 packets with sequence number 0, 1, 2 and 3. The sender an Acknowledgement with acknowledgement number = 2. What is the interpretation of this if the system is using (i) Selective Repeat, (ii) Go Back N protocol?
5. a) Suppose sender has encoded the dataword 1001 using Cyclic Redundancy Check (CRC) where the divisor is from the polynomial of  $x^3+x+1$ . The receiver receives the codeword 1000110. Explain whether the receiver receives the correct codeword or not by using CRC decoder for receiver side.
- b) Suppose you have a 4-bit dataword 1100, and you are using a C(7, 4) Hamming code for error correction. Show the codeword at the sender's side. If the dataword is changed to 1000; show the codeword at the receiver's side. How does the receiver determine the position of the erroneous bit in the codeword?

**Bismillahir Rahmanir Rahim**  
**International Islamic University Chittagong**  
**Department of Computer Science & Engineering**  
*B. Sc. in CSE Semester Final Examination, Autumn-2024*  
**Course Code: CSE-3637    Course Title: Software Engineering**  
**Total marks: 50    Time: 2 hours 30 minutes**

[Answer *all* the questions. Figures in the right hand margin indicate full marks.  
Separate answer script must be used for Group A and Group B]

**Group-A**

CLO    Marks

1    Read the passage and answer the questions 1(a-c):

You are working on the development of a Smart Home Automation System that allows users to control devices such as lights, fans, and air conditioners via a mobile app or voice commands. The system includes features like scheduling devices, monitoring energy consumption, and sending alerts for maintenance needs.

The system will involve interactions with various stakeholders, including the homeowners who use the app, the technicians who install and maintain the devices, and the product managers responsible for defining system features.

- |                                                                                                                                                              |      |   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---|
| a) Identify and categorize at least four stakeholders for the Smart Home Automation System.                                                                  | CLO1 | 3 |
| b) Using the given scenario, draw a viewpoint hierarchy diagram and discuss how it represents different perspectives of the system.                          | CLO3 | 4 |
| c) Based on the scenario, create a use case diagram that helps different stakeholders to better understand the requirements of the system and functionality. | CLO3 | 3 |

OR

- |                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---|
| 1 a) Briefly write about the feasibility study. Now read and follow the given scenario<br>Your retail company is considering starting an in-house delivery service instead of relying on third-party providers. Before making a final decision, you need to conduct a feasibility study.<br>Now as a software engineer, what questions will you rise for a <b>feasibility study</b> in terms of economic, operational and technical? | CLO1 | 4 |
| b) What is use case? For the given scenario find out the key feature and make use cases for those customer interactions with the application.                                                                                                                                                                                                                                                                                        | CLO3 | 6 |

Imagine an **online e-shop** that specializes in selling home appliances, such as refrigerators, washing machines, microwaves, air conditioners, and kitchen gadgets. The e-shop offers a user-friendly platform where customers can browse, compare, and **purchase products**, schedule deliveries, and request installation services. The website also provides detailed product descriptions, user reviews, and real-time customer support.

[Hints: in online e-commerce applications the main feature is product purchasing]



2 Read the passage and answer the questions 2(a-c):

You are tasked with designing a Carpooling Mobile Application that connects drivers offering rides with passengers seeking transportation. The app should include features such as user registration, ride creation, ride booking, fare calculation, real-time location tracking, and payment integration. The design should ensure scalability as the user base grows and incorporate security features to protect user data and payments.

- a) What is the role of abstraction and modularity in designing the Carpooling Mobile Application? Explain how they can help simplify the system design. CLO1 3
- b) Based on the general model of the software design process, outline the key steps you would follow to design the Carpooling Mobile Application. Briefly explain each step. CLO2 4
- c) Define coupling and cohesion in the context of software design. How can you ensure low coupling and high cohesion when designing the modules for ride creation and payment processing? CLO2 3

### Group-B

3 Read the passage and answer the questions 3(a-c):

You are part of a team responsible for maintaining a popular Hospital Management System (HMS) used by hospitals to manage patient records, appointments, billing, and inventory. Recently, you have received multiple requests from hospitals for updates and fixes:

1. A bug in the billing system occasionally calculates incorrect amounts, causing issues for patients.
2. New government regulations require the system to support e-prescriptions for medications.
3. Users have suggested improving the system's performance, as it becomes slow when handling a large number of patients.
4. The system is written in an outdated programming language that is becoming harder to maintain.

- a) Classify the following requests based on the types of software maintenance: CLO2 3
- i. Fixing the bug in the billing system.
  - ii. Adding support for e-prescriptions as per government regulations.
  - iii. Migrating the system from an outdated programming language to a modern one.
- b) For the given scenario, describe the activities involved in software maintenance process. CLO2 4
- c) List and explain at least three factors that would impact the maintenance cost of the Hospital Management System. CLO1 3

- 4 a) You are part of a development team tasked with creating an e-learning platform for a university. The platform will include features such as user registration, course management, assessments, and a grading system. To save time and resources, the team decided to explore software reuse opportunities. CLO2 3

Your task is to identify which type(s) of software reuse can be applied to the following components: **User Registration System, Course Management Module, Grading System**

- b) Draw and explain the software re-engineering process diagram. CLO3 4
- c) Describe the four types of software maintenance and provide an example for each type. CLO3 3
- 5 a) Describe the necessary steps in the COCOMO model. CLO2 5
- b) The fan in and fan out of module X is 3 and 4 respectively. The complexity of the system is 2000. The number of lines in module X is 200, i.e LOC of module X is  $LOC(X) = 200$ . Calculate the structural complexity of module X using card and glass's system complexity. Using combined Henry Kafura's approach and Card glass's approach calculate the data complexity of module X. CLO3 5
- OR
- 5 a) Explain the key process area in the capability maturity model. CLO2 5
- b) Suppose the fan out of a module is: 3, the cumulative number of variables passed to and from the module is: 44, and the complexity of the module is: 39. Use Card and Glass's Systems Complexity in order to find out the structural complexity, data complexity and system complexity of the module. CLO3 5



# International Islamic University Chittagong

Department of Computer Science & Engineering  
B.Sc. in CSE, Final Examination, Autumn 2024

Course Code: CSE-3635

Total Marks: 50

Course Title: Artificial Intelligence

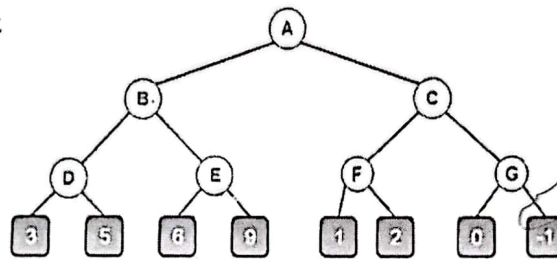
Time: 2 Hours 30 Minutes

[Answer *all* the following questions. Figures in the right hand margin indicate full marks. Use a separate answer script for Group-A and Group-B.]

COs

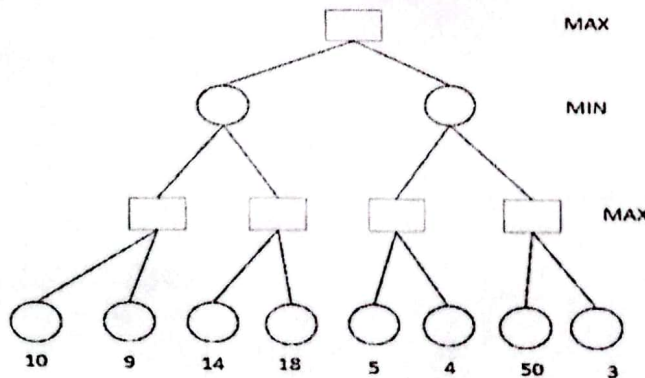
## Group A

1. a) What is CSP? Do you consider map coloring as an example of CSP? If so, identify the variables, domains and constraints associated with this problem. 5 1
- b) What is an imperfect decision in terms of game playing? Consider the following game tree where static scores are all from the first player's point of view. In the tree, which nodes will not be visited using the alpha beta pruning procedure? 5 2



OR

1. a) Consider the Bangladeshi map below. You want to color divisions so that none of their neighbors have the same color. The colors you may choose from are Red, Blue and Green. Formulate the problem. 1
- b) What are the differences between Search-Based Problem and Constraint Satisfaction Problem? Consider the following two player game tree in which the static scores are given from the first player's point of view. Apply the Minimax search algorithm and compute the value of the root of the tree. Also find the most convenient path for Max Node. 5 2



2. a) How can the Wumpus World be described in terms of its features and environment? 2 2
- b) "As per the law, it is academic dishonesty for an IIUC student to cheat in the exam hall. During the AI exam, Rahim, a student of IIUC, has some Cheating materials, and all these materials were provided to him by Rafat, who is also an IIUC student." 8 2  
Now, using forward and backward chaining prove that "Rafat is a dishonest student."



3. a) A survey was conducted at a juice bar, and the following data were collected:

- 8 male students, 5 female students, 3 male teachers and 2 female teachers bought orange juice
- 6 male students, 7 female students, 2 male teachers and 4 female teachers bought mango juice
- 4 male students, 5 female students, 2 male teachers and 2 female teachers bought lemon juice

Answer the following question:

*C = Teacher | Student*

- i. Create a Joint probability distribution table for the given data.
- ii.  $P(C) = ?$
- iii.  $P(C | \text{Female}) = ?$
- iv.  $P(C | \text{Female, Student}) = ?$

Calculate the probability using inference by enumeration.

- b) A well-known symptom of measles is a rash (the event of having which we denote R). Assume that the probability of having a rash if one has measles is  $P(R|M) = 0.95$ . However, occasionally children with flu also develop rash, and the probability of having a rash if one has flu is  $P(R|F) = 0.08$ .

Upon examining the child, the doctor finds a rash. What is the probability that the child has measles?

4. a) What is supervised learning? Briefly explain the concept of mathematical neuron using appropriate diagram.

- b) Consider a single neuron with two inputs  $X_1$  and  $X_2$ , and output  $Y$ , learning the logical OR operation. The truth table is given below.

| Case | $X_1$ | $X_2$ | $Z$ |
|------|-------|-------|-----|
| 1    | 0     | 0     | 0   |
| 2    | 0     | 1     | 1   |
| 3    | 1     | 0     | 1   |
| 4    | 1     | 1     | 1   |

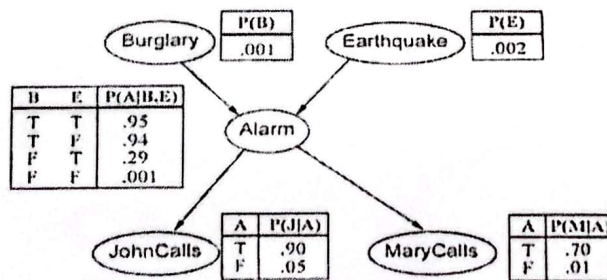
Develop an ANN model using Delta rule, Where,

Threshold,  $T = 0.5$

Learning rate,  $\alpha = 0.2$

Initial weights  $W_1 = 0.1$  and  $W_2 = 0.5$

5. a)



Consider the Bayesian Network above:

Calculate the probability that alarm has sounded, but there is neither a burglary, nor an earthquake occurred, and John and Mary both called.

- b) What is the probability that Mary call?
- c) What is the role of membership function in Fuzzy Logic? Describe the fuzzy logic process diagram in brief.

OR

5. a) Compare top down and bottom up parsing approach in parsing sentence tree.
- b) Convert  $A \Leftrightarrow (B \vee C)$  into CNF. Prove that the contra positive of  $p \rightarrow q$  is equal to  $p \rightarrow q$  using truth table.
- c) What do you mean by uncertainty? What are the different sources of uncertainty.



1(a) OTC

